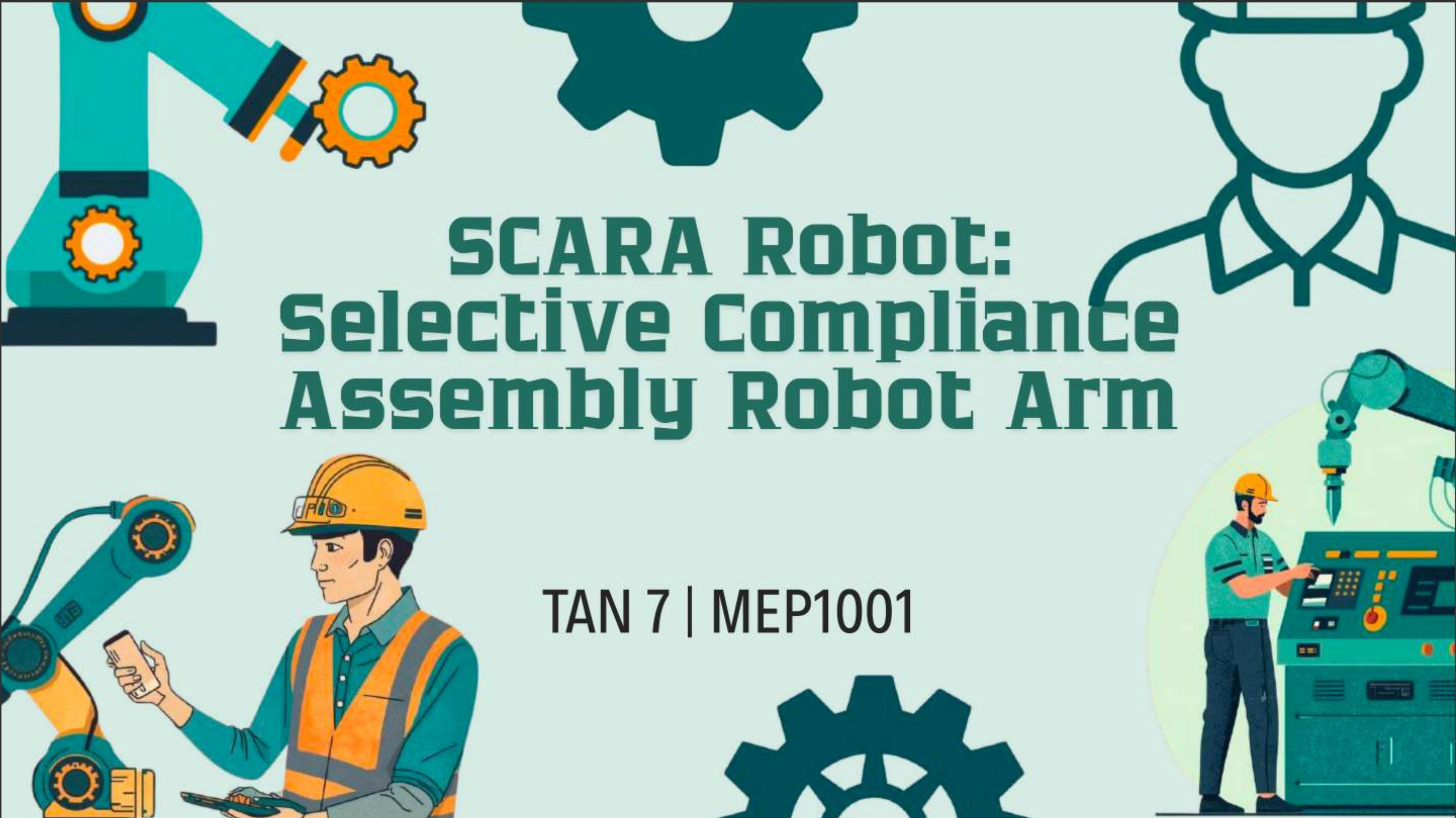


SCARA Robot: Selective Compliance Assembly Robot Arm

TAN 7 | MEP1001



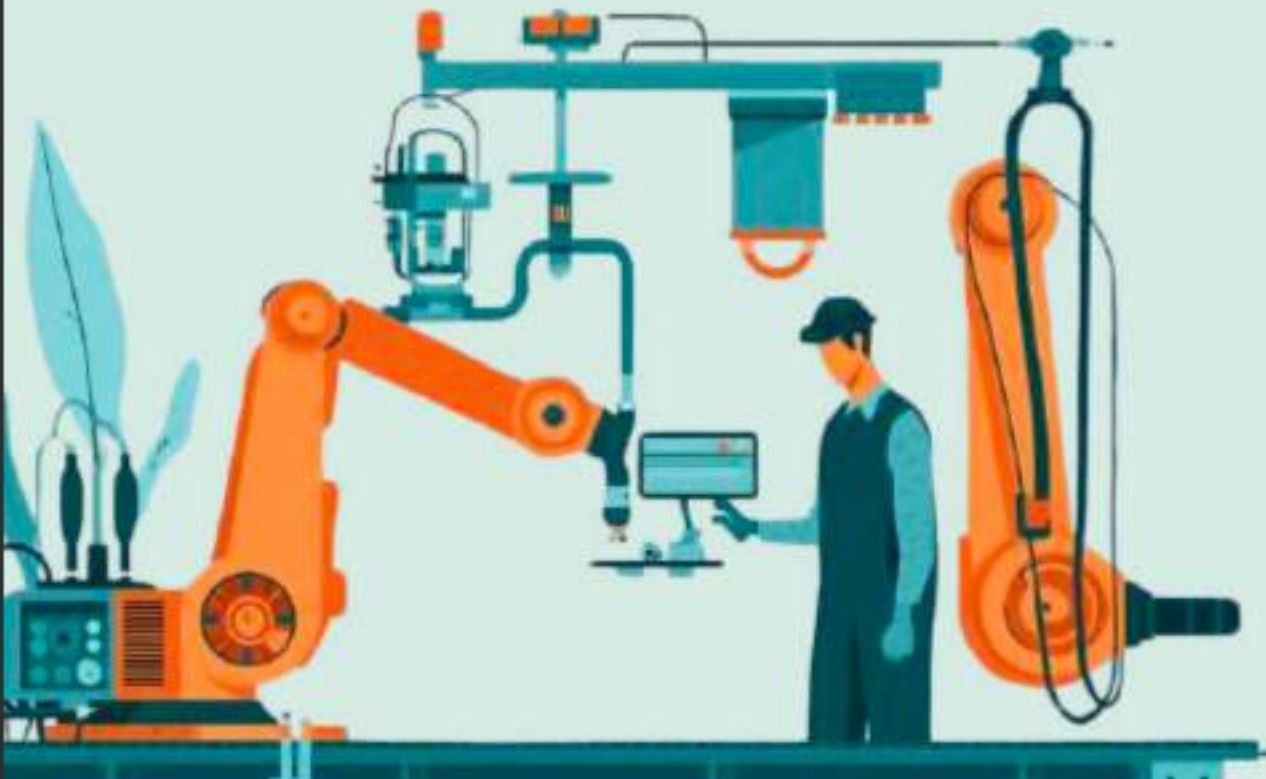


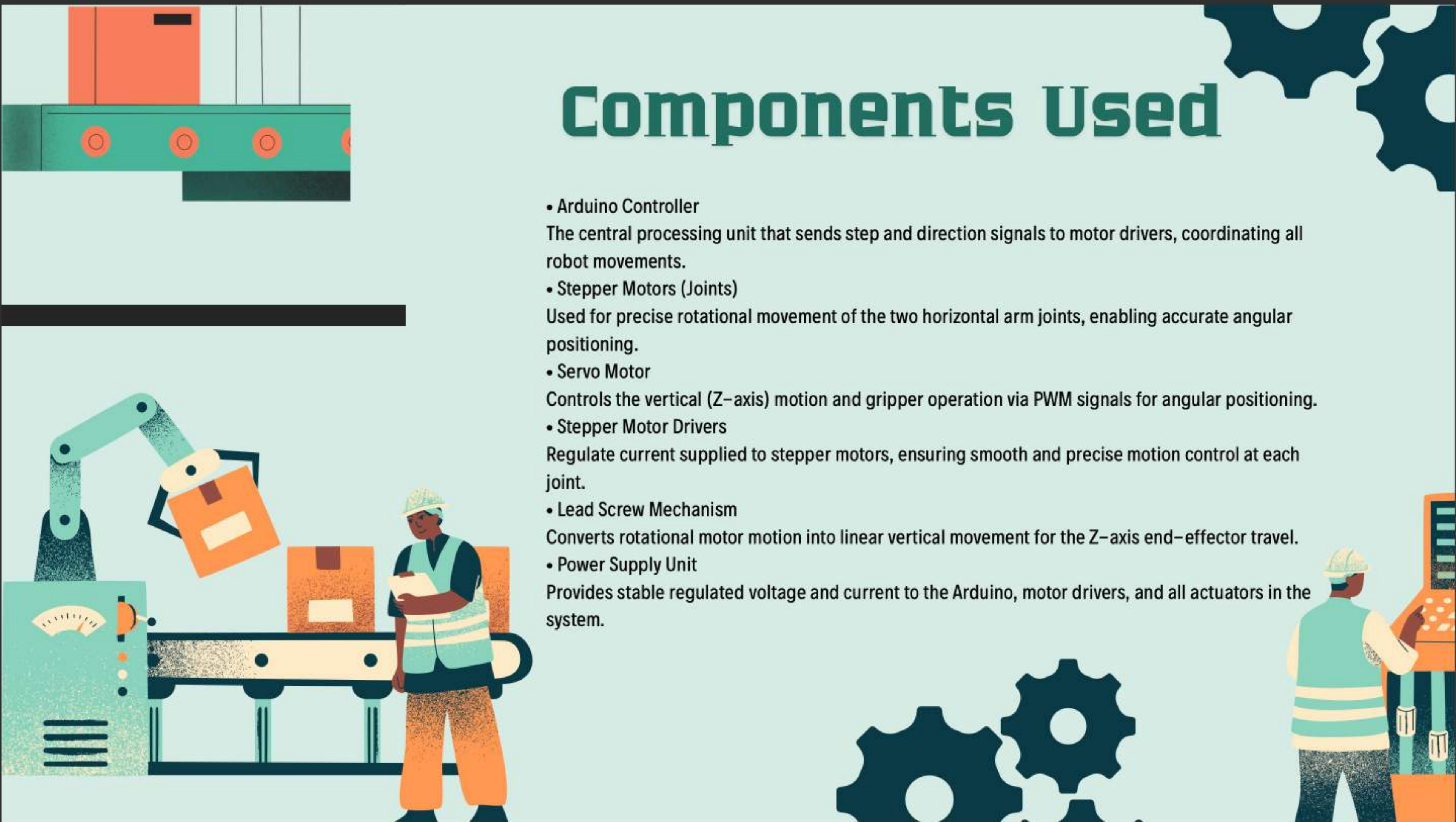
Introduction

A SCARA (Selective Compliance Assembly Robot Arm) robot is designed for fast, precise movement in the horizontal plane. Its unique compliance in the X-Y plane combined with rigidity in the Z-axis makes it ideal for:

- Pick-and-place operations
- Assembly line automation
- High-speed, repeatable positioning

SCARA robots are widely used in industrial manufacturing due to their exceptional speed and accuracy in horizontal tasks.





Components Used

- **Arduino Controller**

The central processing unit that sends step and direction signals to motor drivers, coordinating all robot movements.

- **Stepper Motors (Joints)**

Used for precise rotational movement of the two horizontal arm joints, enabling accurate angular positioning.

- **Servo Motor**

Controls the vertical (Z-axis) motion and gripper operation via PWM signals for angular positioning.

- **Stepper Motor Drivers**

Regulate current supplied to stepper motors, ensuring smooth and precise motion control at each joint.

- **Lead Screw Mechanism**

Converts rotational motor motion into linear vertical movement for the Z-axis end-effector travel.

- **Power Supply Unit**

Provides stable regulated voltage and current to the Arduino, motor drivers, and all actuators in the system.

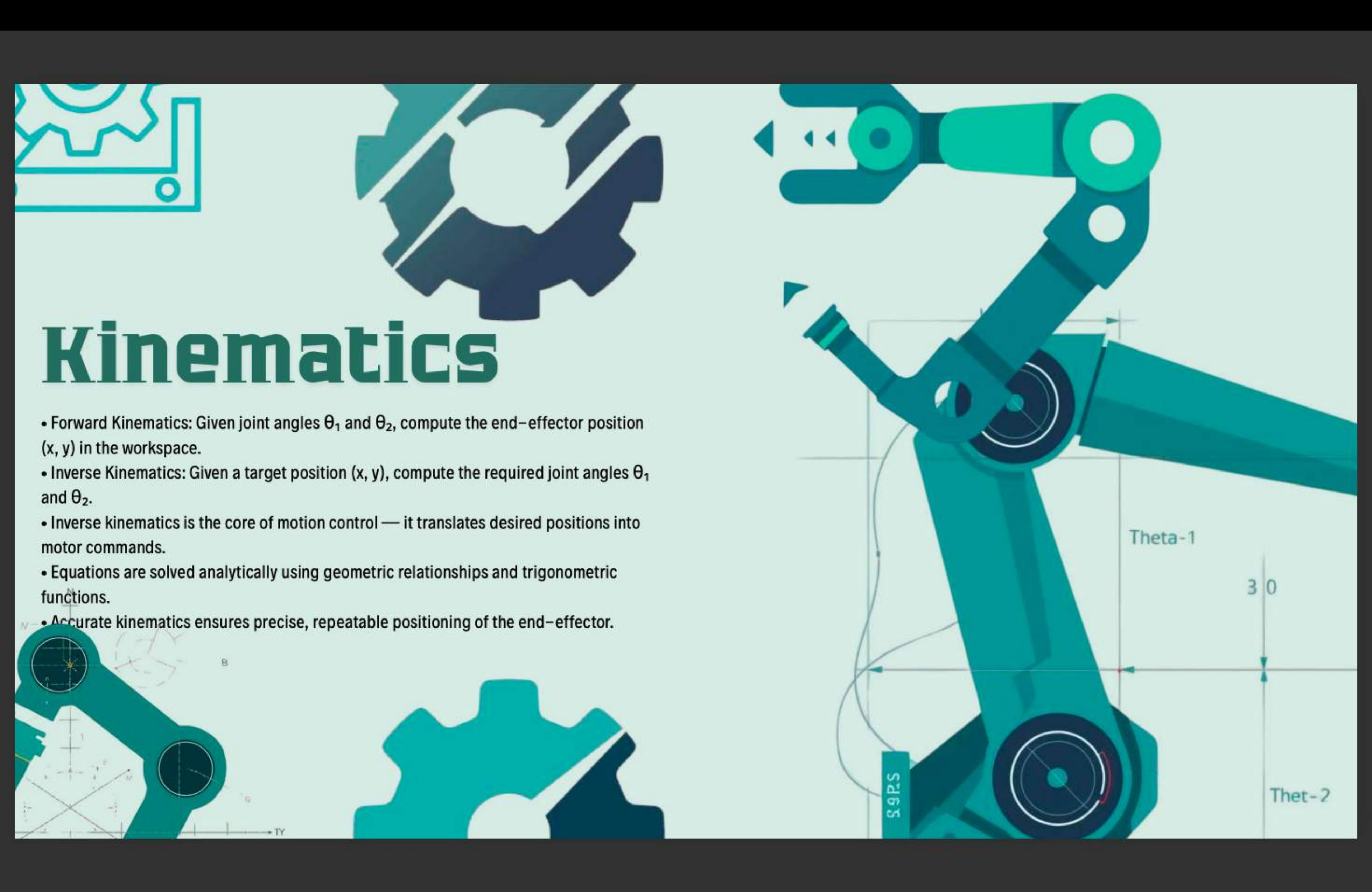


Working Principle

The SCARA robot operates through a structured motion control pipeline. The user inputs the desired end effector position in Cartesian coordinates. An inverse kinematics algorithm then computes the required joint angles for each arm segment to reach that target position accurately.

The Arduino controller processes the computed joint angles and sends step/direction signals to the stepper motor drivers. The drivers regulate current to actuate each motor, rotating the arm segments to the target position enabling fast, repeatable, and precise horizontal plane movement.



The background of the slide features a stylized illustration of a robotic arm in shades of teal and dark blue. The arm is positioned on the right side, with its joints and segments clearly visible. In the upper left corner, there are two interlocking gears, one in a lighter teal and one in a darker blue. The overall aesthetic is clean and technical, with a light teal background.

Kinematics

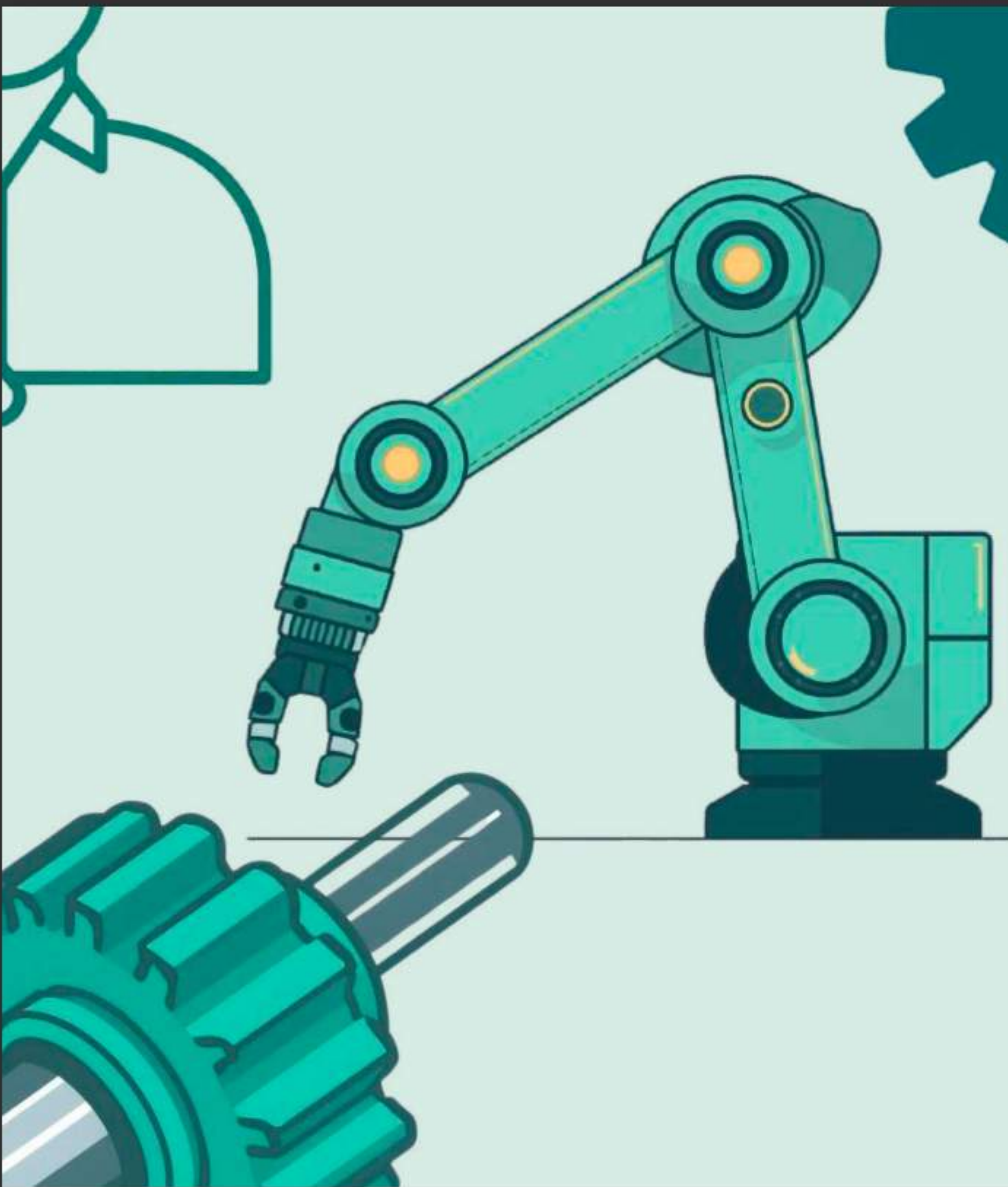
- Forward Kinematics: Given joint angles θ_1 and θ_2 , compute the end-effector position (x, y) in the workspace.
- Inverse Kinematics: Given a target position (x, y) , compute the required joint angles θ_1 and θ_2 .
- Inverse kinematics is the core of motion control — it translates desired positions into motor commands.
- Equations are solved analytically using geometric relationships and trigonometric functions.
- Accurate kinematics ensures precise, repeatable positioning of the end-effector.

Theta-1

3 0

Thet-2

S9PS

An illustration of a teal robotic arm with two joints, each marked with a yellow dot. The arm is positioned on the left side of the slide. In the background, there are several large teal gears. A small teal gear is at the top, a large one is at the bottom left, and another is at the bottom center. A thin teal line outlines a shape in the top left corner. On the right side, there is a partial illustration of a person wearing a teal hard hat and a teal shirt, holding a small teal object.

Mechanical Design

- Two rotational arms enabling precise horizontal (X-Y plane) movement
- Vertical Z-axis for controlled up/down motion via lead screw
- Base constructed from wood for vibration damping, covered with sheet metal for structural strength
- 3D printed components used for joint housings and motor mounts
- Compact, rigid frame designed for repeatability and stable operation

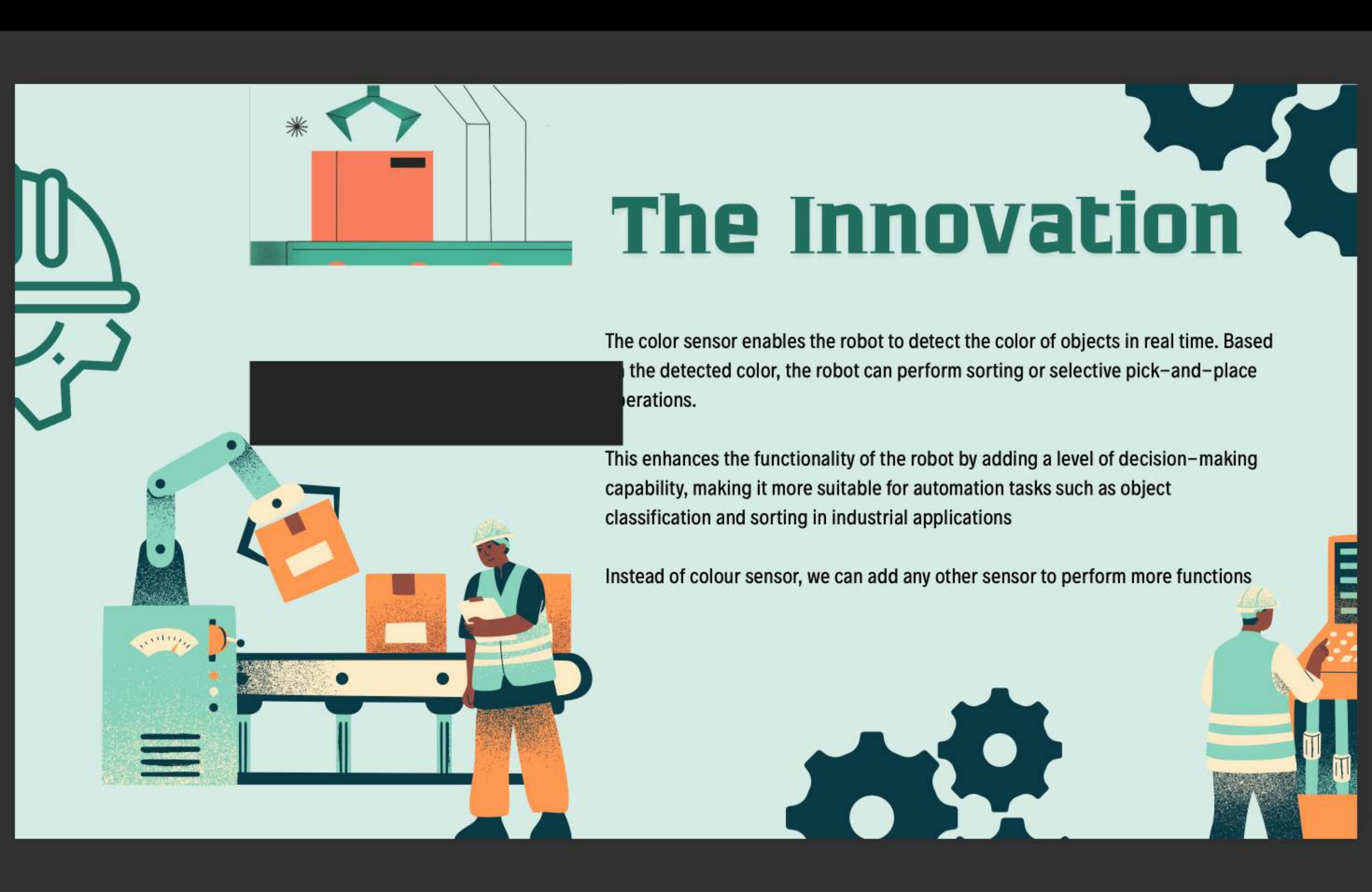


Electronics and Wiring

The Arduino microcontroller serves as the central hub of the electrical system, sending step and direction signals to the stepper motor drivers. Each driver receives precise pulse sequences that determine the speed, direction, and number of steps each motor executes. The drivers regulate motor current to ensure smooth, accurate, and consistent actuation across all joints.

Signal wiring uses lightweight jumper wires to carry low-current control signals between the Arduino and motor drivers. Power lines use thicker gauge wires to safely handle the higher current demands of the stepper and servo motors. Proper wire management and secure connections are essential for reliable and repeatable robot performance.





The Innovation

The color sensor enables the robot to detect the color of objects in real time. Based on the detected color, the robot can perform sorting or selective pick-and-place operations.

This enhances the functionality of the robot by adding a level of decision-making capability, making it more suitable for automation tasks such as object classification and sorting in industrial applications

Instead of colour sensor, we can add any other sensor to perform more functions



Summary



SCARA Robot concept introduced as a fast, precise robotic arm for horizontal plane movement. Project objective focused on designing a robot capable of accurate positioning and repeatable pick-and-place tasks with efficient motion control.

Mechanical build utilized two rotational arms, a vertical axis, wood base with sheet metal cover, and 3D printed joints. Electronics centered on Arduino with stepper motor drivers, servo motor, and lead screw for linear motion.



Control system used step/direction pulses for stepper motors and PWM signals for the servo. Robot achieved repeatable movements and basic pick-and-place tasks. Key challenges include lack of feedback and 3D printed part limitations. Future work targets closed-loop control and metal components for improved durability and precision.



The background is a light teal color with various industrial illustrations. At the top left, a green conveyor belt carries an orange box towards a green robotic arm. At the top right, there's a large green gear and a stylized green outline of a hard hat. In the bottom left, a worker in a green vest and yellow hard hat stands next to a conveyor belt with orange boxes, holding a clipboard. In the bottom right, another worker in a green vest and yellow hard hat stands at a control panel with a screen and buttons. Several large green gears are scattered throughout the background.

Thank You!